Supporting Information

TriACP: An ESM-2-Enhanced Multistage Framework for Anticancer Peptide Identification, Cancer-Type Activity Prediction, and Potency Classification

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Supplementary TABLE S1

TABLE S1
TABULAR ILLUSTRATION OF EXISTING STUDIES ON TRIACP.

Classifier Name	Operating Algorithm	Publication Year
Multi-ACPNet [18]	BiLSTM, CNN & GCN	2026
DeepACPred [22]	ESM2–DNN	2026
ACP-CLB [17]	CNN, BiLSTM, ProtBERT + Conv1D	2025
DUO-ACP [15]	Pre-training–based model	2025
ACP-CapsPred [16]	Capsule Network	2024
Contrastive Learning [12]	Contrastive Learning enhanced CNN	2024
ACP-PDAFF [10]	Pre-training–based model (ProtBert)	2024
MA-PEP [11]	Attention-based multimodal	2024
mACPpred2.0 [13]	Stacked Deep Learning (SDL)	2024
UniDL4BioPep [8]	PLM–CNN	2023
ACP-MLC [14]	Random Forest	2023
GRDF [7]	Deep Forest	2023
TriNet [9]	CNN, Bi-LSTM & Transformer encoder	2023
ACP-Check [6]	BiLSTM	2022
iACP-DRLF [19]	LSTM + LightGBM	2021

Supplementary TABLE S2

TABLE S2

DISTRIBUTION OF TRAINING AND TESTING ON SET 2 AND SET 3.

Dataset	Train/Test	Positive	Negative	Total
Set 2	Train dataset	716	677	1393
	Test dataset	186	170	356
Set 3	Train dataset	242	242	484
	Test dataset	61	61	122

Supplementary TABLE S3

TABLE S3
DISTRIBUTION OF THE STAGE-II DATASET USED FOR ONE-VS-ALL CANCER-TYPE PREDICTION

Cancer Types	No. of ACPs	Training Positive/Negative	Testing Positive/Negative
Colon	137	109/59	28/15
Breast	124	99/69	25/18
Cervix	105	84/84	21/22
Skin	106	84/84	22/21
Lung	94	75/93	19/24
Prostate	89	71/97	18/25
Blood	63	50/118	13/30

Supplementary TABLE S4

TABLE S4

PERFORMANCE COMPARISON OF THE PROPOSED STAGE-I MODEL WITH EXISTING ACP PREDICTION METHODS ON SET 2 AND SET 3.

Dataset	Methodology	ACC	Precision	Recall	Specificity	F1-Score	MCC
Set 2	ACP-Check	0.93	0.93	0.93	0.93	0.93	0.86
	GRDF	0.9460	0.9769	0.9086	0.9774	0.9415	0.8856
	UniDL4BioPep	0.946	0.9172	**0.978**	0.918	0.9463	0.894
	TriNet	0.8770	0.9080	0.8880	0.8830	0.8970	0.7600
	ACP-PDAFF	0.94	0.96	0.92	0.96	0.94	0.88
	MA-PEP	0.9356	0.9490	0.9227	0.9485	0.9356	0.8714
	Contrastive Learning (CL)	0.9381	0.9775	0.8969	0.9794	-	0.8793
	mACPpred2.0	0.857	0.575	0.807	0.868	0.671	0.597
	ACP-CLB	0.9533	0.9474	0.96	0.9467	0.9536	0.9067
	ACP-CapsPred	0.9571	0.9598	0.9543	0.9593	**0.9590**	0.9127
	Multi-ACPNet	0.9536	-	0.9124	**0.9948**	-	0.9103
	TriACP*	**0.9579**	**0.9777**	0.9409	0.9765	0.9589	**0.9164**
Set 3	ACP-Check	0.78	0.77	0.80	0.77	0.78	0.56
	GRDF	0.7712	0.7683	0.7826	0.7595	0.7754	0.5454
	UniDL4BioPep	0.735	0.7366	0.734	0.737	0.7352	0.471
	ACP-PDAFF	0.8	0.77	0.83	0.76	0.8	0.6
	MA-PEP	0.8081	0.7509	0.8779	0.7384	0.81	0.6224
	Contrastive Learning (CL)	0.8081	0.8118	0.8023	0.814	-	0.6163
	ACP-MLC	0.7116	0.7032	0.8330	0.6392	0.7627	-
	DUO-ACP	0.828	-	0.885	**0.9020**	-	0.663
	ACP-CapsPred	0.8025	**0.8170**	78.12	0.8239	79.86	0.5952
	ACP-CLB	0.8110	0.7744	0.8778	0.7442	0.8229	0.6277
	Multi-ACPNet	0.8140	-	0.8214	0.8021	-	0.6283
	TriACP*	**0.8361**	0.7808	**0.9344**	0.7377	**0.8508**	**0.6855**

Supplementary TABLE S5

TABLE S5

PERFORMANCE COMPARISON OF THE PROPOSED STAGE-II MODEL WITH EXISTING METHODS FOR CANCER-TYPE ACTIVITY PREDICTION USING 10-FOLD CROSS-VALIDATION.

Cancer type	Method	ACC	SEN	SP	F1-score	AUC	MCC	Hamming loss
Colon	Multi-ACPNet	0.8690	**0.8967**	0.8129	0.8948	0.8716	0.7206	-
	Duo-ACP	0.791	0.825	0.662	0.843	0.834	0.535	-
	ACP-MLC	0.818	0.846	0.778	0.846	0.923	0.624	-
	Proposed Model	**0.8760**	0.8810	**0.8667**	**0.9024**	**0.9350**	**0.7342**	-
Breast	Multi-ACPNet	0.8557	0.8859	0.8154	0.8770	0.9065	0.7088	-
	Duo-ACP	0.825	0.838	0.759	0.854	0.9	0.636	-
	ACP-MLC	0.727	0.700	0.750	0.7000	**0.925**	0.450	-
	Proposed Model	**0.8811**	**0.8981**	**0.8443**	**0.9119**	0.9189	**0.7303**	-
Cervix	Multi-ACPNet	**0.8881**	0.8364	**0.9416**	**0.8760**	**0.9115**	**0.7839**	-
	Duo-ACP	0.853	0.856	0.858	0.852	0.91	0.706	-
	ACP-MLC	0.636	0.545	0.727	0.6000	0.802	0.277	-
	Proposed Model	0.8398	**0.8614**	0.8235	0.8218	0.8830	0.6792	-
Skin	Multi-ACPNet	0.8912	0.8744	0.9114	0.8736	0.9231	0.7937	-
	Duo-ACP	0.863	0.86	0.857	0.864	0.903	0.725	-
	ACP-MLC	0.818	0.429	**1.00**	0.600	0.876	0.582	-
	Proposed Model	**0.9121**	**0.9005**	0.9261	**0.9179**	**0.9566**	**0.8242**	-
Lung	Multi-ACPNet	0.8592	0.8609	0.8604	0.8585	0.9433	0.7336	-
	Duo-ACP	0.858	0.827	0.855	0.844	0.905	0.714	-
	ACP-MLC	0.864	0.667	**0.938**	0.727	0.917	0.642	-
	Proposed Model	**0.8837**	**0.8852**	**0.8852**	**0.8916**	**0.9496**	**0.7663**	-
Prostate	Multi-ACPNet	0.8909	0.8528	0.9199	0.8670	**0.9399**	**0.7818**	-
	Duo-ACP	**0.891**	**0.875**	0.91	**0.87**	0.923	0.776	-
	ACP-MLC	0.864	0.864	1.**00**	0.727	0.886	0.690	-
	Proposed Model	0.8708	0.8611	0.8792	0.8611	0.9396	0.7403	-
Blood	Multi-ACPNet	0.8338	0.6524	0.9094	0.6838	0.8302	0.5935	-
	Duo-ACP	0.763	0.603	0.831	0.603	0.825	0.434	-
	ACP-MLC	0.682	0.500	0.786	0.533	0.750	0.295	-
	Proposed Model	**0.9225**	**0.9018**	**0.9309**	**0.8707**	**0.9614**	**0.8164**	-
Average	Multi-ACPNet	0.8697	0.8371	**0.8816**	0.8472	0.9033	0.7309	0.1303
	Duo-ACP	0.8350	0.8120	0.8190	0.8190	0.8860	0.6470	0.1650
	ACP-MLC	0.7730	0.6080	0.8540	0.6760	0.8680	0.5090	0.1570
	Proposed Model	**0.8837**	**0.8842**	0.8790	**0.8825**	**0.9349**	**0.7558**	**0.1163**

Supplementary TABLE S6

TABLE S6
ABLATION PERFORMANCE OF THE PROPOSED STAGE-I MODEL ON SET 2 AND SET 3 AFTER REMOVAL OF INDIVIDUAL MODEL COMPONENTS.

Dataset	Methodology	ACC	Precision	Recall	F1-Score	MCC	FPR
Set 2	Without AAindex	0.9410	0.9769	0.9086	0.9415	0.8845	0.0235
	Without BLOSUM62	0.9522	0.9774	0.9301	0.9532	0.9057	0.0235
	Without Pre-trained Model	0.8904	0.8808	0.9140	0.8971	0.7807	0.1353
	Without Capsule Network	0.9494	**0.9884**	0.9140	0.9497	0.9018	**0.0117**
	Without MLP	0.9354	0.9503	0.9247	0.9373	0.8710	0.0529
	TriACP	**0.9579**	0.9777	**0.9409**	**0.9589**	**93.59%**	0.0235
Set 3	Without AAindex	0.8033	0.7403	**0.9344**	0.8261	0.6286	0.3279
	Without BLOSUM62	0.7951	0.7647	0.8525	0.8062	0.5941	0.2623
	Without Pre-trained Model	0.7787	**0.8542**	0.6721	0.7523	0.5705	**0.1148**
	Without Capsule Network	0.7951	0.7308	**0.9344**	0.8201	0.6145	0.3443
	Without MLP	0.8115	0.75	**0.9344**	0.8321	0.6427	0.3115
	TriACP	**0.8361**	0.7808	**0.9344**	**0.8508**	**0.6855**	0.2623

Supplementary TABLE S7

TABLE S7

PERFORMANCE COMPARISON OF DIFFERENT MLP DIMENSIONALITY SETTINGS FOR THE HANDCRAFTED FEATURE BRANCH ON STAGE-I ACP IDENTIFICATION USING SET 2 AND SET 3.

Dataset	MLP Dimensions	ACC	Precision	Recall	F1-Score	MCC	AUC
Set 2	mlp32	0.9466	0.9514	**0.9462**	0.9488	0.8931	0.9828
	mlp64	0.9438	0.9511	0.9409	0.9460	0.8875	0.9820
	mlp128 (Proposed)	**0.9579**	**0.9776**	0.9409	**0.9589**	**0.9164**	**0.9851**
	mlp256	0.9438	0.9560	0.9355	0.9456	0.8878	0.9826
Set 3	mlp32	0.8033	0.7342	**0.9508**	0.8286	0.6348	**0.9213**
	mlp64	0.8033	0.7342	**0.9508**	0.8286	0.6348	0.8992
	mlp128 (Proposed)	**0.8361**	0.7808	0.9344	**0.8507**	**0.6855**	0.8995
	mlp256	0.8279	0.7858	0.9016	0.8397	0.6630	0.9008

Supplementary TABLE S8

TABLE S8

PERFORMANCE COMPARISON OF DIFFERENT MLP PROJECTION DIMENSIONS FOR HANDCRAFTED FEATURE COMPRESSION AVERAGE RESULTS ON STAGE-II CANCER-TYPE ACTIVITY PREDICTION.

Method	MLP Dimensions	ACC	Precision	Recall	F1-Score	MCC	AUC
TriACP	mlp32	0.8612	0.8532	0.8933	0.8630	0.7166	**0.9417**
	mlp64	0.8764	0.8719	**0.8973**	0.8784	0.7467	0.9409
	mlp128 (Proposed)	**0.8837**	**0.8819**	0.8842	**0.8825**	**0.7558**	0.9349
	mlp256	0.8652	0.8713	0.8773	0.8615	0.7300	0.9404

Supplementary Fig. S1

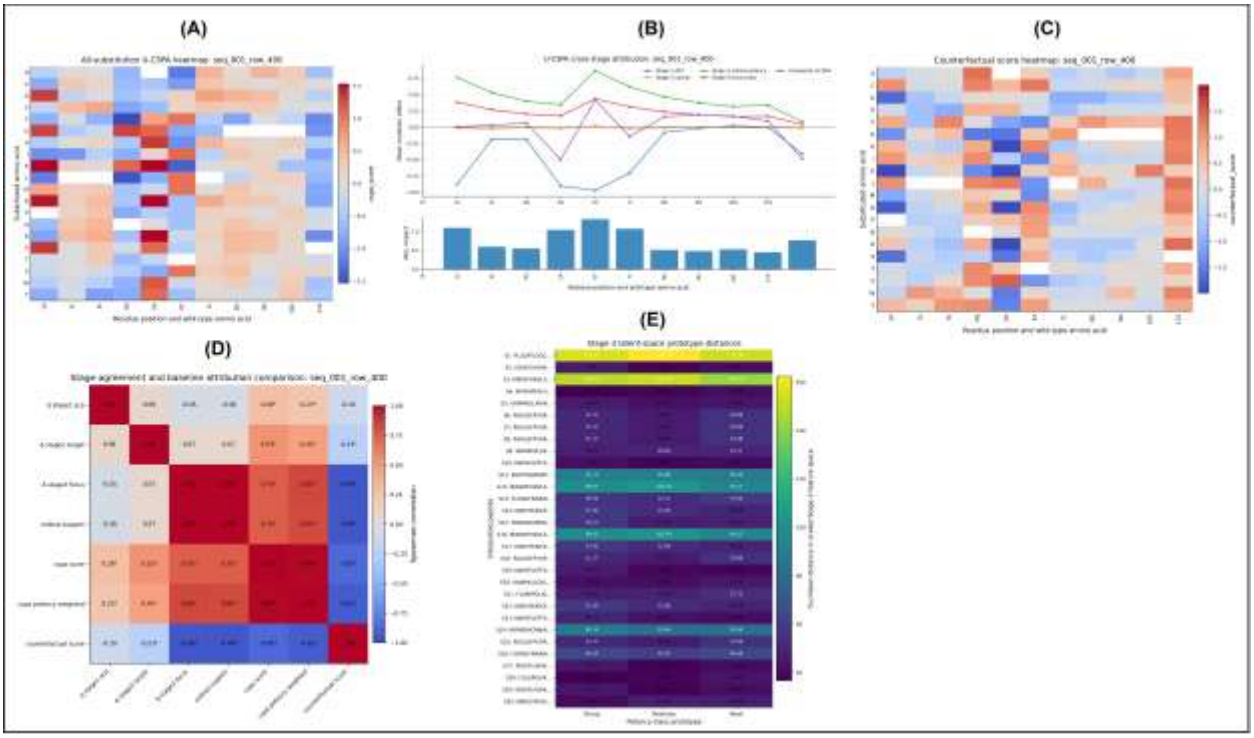


Figure S1. Sequence-level interpretability analysis of the prediction framework. (A) All-substitution U-CSPA heatmap showing residue-wise mutation sensitivity. (B) Cross-stage attribution profiles showing residue contributions across prediction stages. (C) Counterfactual mutation score heatmap showing predicted effects of individual substitutions. (D) Agreement analysis among stage-specific attribution measures and mutation-based scores. (E) Stage-III latent-space prototype distance analysis showing peptide distances to potency-class prototypes in the learned feature space.